

Data Analytics & Business Intelligence

Task-1 Final Project Report

This project focuses on Data Cleaning, Exploratory Data Analysis (EDA), SQL analysis, and Dashboard Reporting using the Superstore dataset. The objective is to analyze business sales and profit performance using Excel and Power BI.

1. Introduction

The Superstore dataset was used for business analysis. The project includes data cleaning, pivot table analysis, SQL queries, and dashboard creation.

2. Data Cleaning

The dataset was cleaned by removing duplicate values and handling missing data. Date columns were corrected and formatted properly for analysis.

3. Excel Analysis

Pivot tables and charts were created in Excel to analyze category-wise sales, region-wise profit, and sales trends over time.

4. SQL Queries Used

1. Total Sales by Category

```
SELECT Category, SUM(Sales) AS Total_Sales FROM Superstore GROUP BY Category;
```

2. Total Profit by Region

```
SELECT Region, SUM(Profit) AS Total_Profit FROM Superstore GROUP BY Region;
```

3. Total Orders Count

```
SELECT COUNT(Order_ID) AS Total_Orders FROM Superstore;
```

5. Dashboard Analysis

A Power BI dashboard was designed using KPI cards, bar charts, pie charts, and line charts to visualize sales and profit performance.

6. Business Insights

- Technology category generated the highest sales.
- West region contributed the highest profit.
- Sales increased steadily over the years.
- Consumer segment generated maximum sales.
- Office Supplies category showed stable performance.

7. Conclusion

This project successfully demonstrated data cleaning, analysis, and visualization techniques using Excel, SQL, and Power BI.

8. Dashboard Preview

